

Simulating Central Vision Loss for Multisensory Navigation in Virtual Environments

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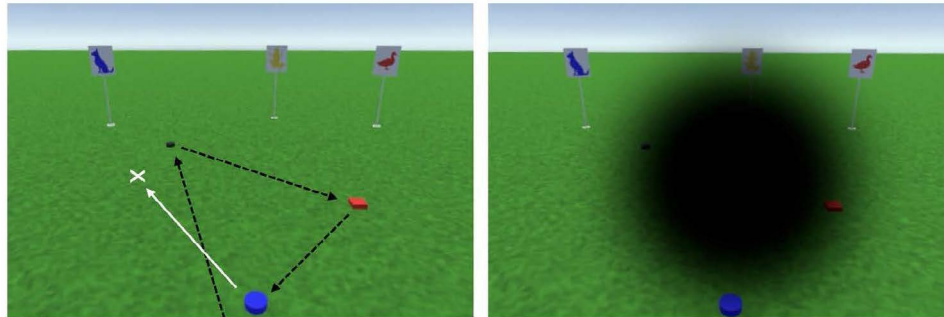


Figure 1: An example trial of the navigation homing task in the no scotoma condition with the encoding segments in dashed lines and response segment in the solid line (left) and the same trial with the gaze-contingent central scotoma condition (right).

ABSTRACT

Low vision is a prevalent health issue that restricts individuals' accessibility to virtual reality technology due to reduced visual input. In this study, we developed a low vision simulation to model central vision loss, a symptom of eye conditions like macular degeneration. Participants completed a navigation task using both visual and self-motion cues under normal and simulated low vision conditions. Results indicate that simulated central vision loss alters sensory cue use in navigation. These preliminary findings highlight the potential of low vision simulations for future research, including simulating different types or amounts of vision loss and aligning simulations with the experiences of people living with low vision.

Index Terms: Low vision, virtual environments, spatial navigation, multisensory integration.

1 INTRODUCTION

There are approximately 217 million people worldwide living with moderate to severe low vision – uncorrectable vision loss that affects daily functioning [4]. Low vision, which can be caused by conditions such as macular degeneration, often changes how sensory cues are used during navigation [1]. Research on how people with low vision interact with virtual environments (VE) is in its early stages [6] and multisensory integration for navigation has not been the focus. Understanding multisensory-cue use with low vision in VEs is critical as virtual reality (VR) becomes increasingly integral to education, rehabilitation, and other applications, requiring accessibility for all users.

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Navigating both real and virtual environments relies on a variety of sensory spatial cues, including visual landmarks and non-visual self-motion cues (e.g., proprioception and vestibular inputs), which provide information about acceleration and changes in direction. Having multiple sensory cues is particularly important in VEs, as single-sensory cue performance tends to be more variable in VR than real-world settings [2]. This variability underscores the need to understand how multisensory integration supports navigation in VEs, particularly for individuals with low vision.

However, not all low vision pathologies manifest in the same way and experiences can vary significantly between individuals. Given the prevalence and complexity of low vision pathologies, researchers need effective methods to investigate ways to improve VR accessibility, especially when obtaining large samples of low vision participants is challenging. Here, we present the use of low vision simulation in VR to study how sensory integration in navigation may change when experiencing central vision loss.

2 METHOD

Ten participants ($M_{age} = 18.9$; 6 female, 4 male) with at least 20/40 or corrected-to-20/40 visual acuity (tested using a Snellen chart) completed a navigation homing task in immersive VR under two blocked conditions in a randomized order: (1) no scotoma and (2) central scotoma. The task began with an encoding phase where participants walked three path segments, marked by discs on the ground (first black, then red, and blue last) with three landmarks present (animal signs; Figure 1). At the blue target, the screen went black for 10s and one of four sensory cue manipulations occurred¹: (1) vision only: participants were spun in a swivel chair to disable previously encoded self-motion cues, (2) self-motion cues only: the signs in the environment disappeared, (3) two consistent cues: no manipulation (self-motion and vision signal same location for black target), or (4) two conflicting cues: the signs shifted 15° (self-motion and vision signal different locations for black target).

¹Participants completed 64 trials, with 8 repetitions for each of the 2x4 unique variable combinations. The disc locations changed for each trial.

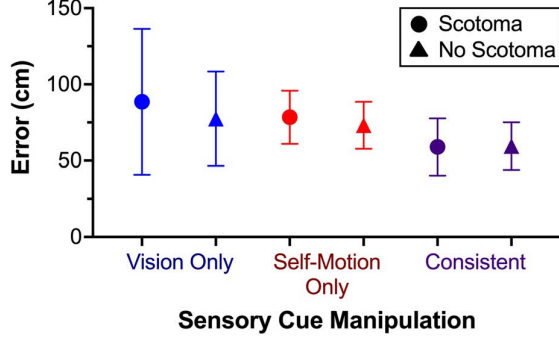


Figure 2: Preliminary results for accuracy (reported as error) of homing performance. Error bars represent SD. Conflicting cues are not presented as the spatial conflict renders accuracy irrelevant due to the absence of a singular correct target location.

Then, in the response phase, participants walked to where they remembered the black disc (without any discs present).

Participants viewed the VE (Figure 1) through a wireless HTC VIVE Pro Eye head-mounted display (HMD). This HMD has built-in eye-tracking with a 110° trackable field of view at a rate of 120Hz. The built-in eye-tracking allowed for a gaze-contingent central scotoma—an area of vision loss that occurs in the central field of view. The scotoma size was approximately 20° at the average eye-to-screen distance in the HMD.

3 RESULTS & DISCUSSION

Navigation accuracy was measured as the difference between the participant's response position in the response phase and the actual black disc location, reported as error in cm. Given the small sample, we present results descriptively (Figure 2). First, error with the central scotoma ($M = 88.6$, $SD = 47.8$) was larger than with no scotoma ($M = 77.5$, $SD = 30.9$) in the vision-only manipulation. However, there were no differences between scotoma/no scotoma in either the self-motion cues (scotoma: $M = 83.9$, $SD = 17.4$; no scotoma: $M = 84.3$, $SD = 16.9$) or the consistent cues manipulation (scotoma: $M = 59.0$, $SD = 18.7$; no scotoma: $M = 59.6$, $SD = 15.6$). These results suggest that a central scotoma may affect navigational accuracy when relying on visual landmarks alone, but it has little impact when additional sensory information is available. This finding highlights the potential for multisensory integration to mitigate navigation challenges in simulated low vision conditions.

The conflicting cue manipulation can be used to calculate the perceptual importance or weighting of visual cues compared to self-motion cues (w_v ; see [3] for the derivation of the formula). The following formula examines how much each participant's response location shifts with the spatial conflict compared to locations in the self-motion (d_{sm}) and visual (d_v) cue manipulations:

$$w_v = \frac{1/d_v^2}{1/d_v^2 + 1/d_{sm}^2} = \frac{d_{sm}^2}{d_{sm}^2 + d_v^2} \quad (1)$$

A visual weight greater than 0.5 represents an observer who relies on visual landmark information more than self-motion. Preliminary results showed that both the no scotoma ($M = 0.62$, $SD = 0.16$) and scotoma ($M = 0.61$, $SD = 0.31$) conditions had similar visual weights, on average. However, the inter-individual variability appears to be much larger in the scotoma condition (Figure 3). The greater variability points to individual differences in how participants adapt to reduced visual input, potentially reflecting varied sensory integration strategies. Understanding this variability in

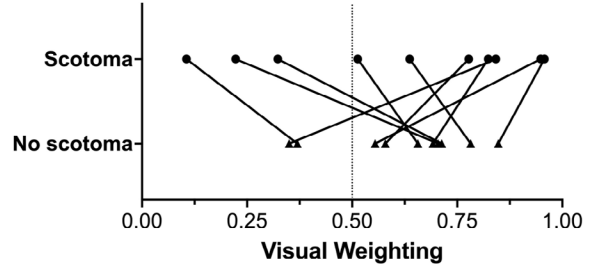


Figure 3: Preliminary visual weighting (w_v) results. The vertical dotted line represents equal weighting of visual and self-motion cues.

simulated low vision may provide insights into how individuals living with low vision adapt to similar reduced vision.

4 CONCLUSIONS & FUTURE DIRECTIONS

These preliminary results suggest that simulated central vision loss may impact both navigational performance and sensory integration. While VR-based simulations themselves are not novel, their application to studying sensory integration strategies represents a significant advancement, as it provides critical insights into how individuals utilize self-motion cues in response to altered visual information. Although still collecting data, this research has important implications for developing potential methods to improve the accessibility of VEs. Supporting more effective and accessible navigation methods requires designing systems that move beyond traditional navigation interfaces (e.g., joysticks, teleportation, walking-in-place) to ensure accurate encoding of self-motion cues.

While simulations are a tool for controlled experimentation, they cannot fully replicate the lived experiences of individuals with low vision at the neural or behavioral level [5]. Therefore, the long-term goal of this research is to compare the navigation performance and sensory integration with simulated low vision to individuals with low vision to test the generalization of this simulation's findings. By bridging the gap between simulation and real-world experience, this research aims to understand of how sensory cues are used in navigation and inform the development of accessible VR systems that are better tailored to the needs of individuals with low vision.

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